RIGHT DERIVED FUNCTORS OF THE INCLUSION OF SHEAVES INTO PRESHEAVES ARE THE COHOMOLOGY PRESHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. The author continues with a lower-case 'which' after the displayed formula (source line 1812); that capitalization is reproduced verbatim and no wording was normalised.

Editorial note. The label lemma-include-modules occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference, which occurs inside the counted statement, is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Inclusion functor from sheaves of modules to presheaves of modules on a ringed site. Same chapter, the forgetful functor i : Mod(C) -> PMod(C) and its left exactness. Supplies the functor whose right derived functors are computed.
- Injective sheaf of O-modules and injective resolutions. Same chapter, injective objects of Mod(O) and the existence of injective resolutions. Supplies the resolution F -> I^bullet from which the derived functors are computed.
- Cohomology H^p(U, F) computed from an injective resolution. Same chapter, the definition of H^p(U, F) as the p-th cohomology of the section complex Gamma(U, I^bullet). Supplies the identification of the displayed kernel modulo image with H^p(U, F).

Source reference. Original section 01FU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 601–601. Same-chapter navigation pointer occurring inside the counted statement at source line 1797, where the author refers the reader to the discussion in a designated section. The pointer only resolves the reference for the reader; the discussion is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Consider the functor $i : Mod(\mathcal{C}) \to PMod(\mathcal{C})$. It is a left exact functor with right derived functors given by*

$$R^p i(\mathcal{F}) = \underline{H}^p(\mathcal{F}) : U \longmapsto H^p(U, \mathcal{F})$$

see discussion in Section 01FU.

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Proof. It is clear that i is left exact. Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in $Mod(\mathcal{O})$. By definition $R^p i$ is the pth cohomology *presheaf* of the complex $\mathcal{I}^\bullet$. In other words, the sections of $R^p i(\mathcal{F})$ over an object U of $\mathcal{C}$ are given by

$$\frac{\mathrm{Ker}(\mathcal{I}^n(U) \to \mathcal{I}^{n+1}(U))}{\mathrm{Im}(\mathcal{I}^{n-1}(U) \to \mathcal{I}^n(U))}.$$

which is the definition of $H^p(U, \mathcal{F})$. □